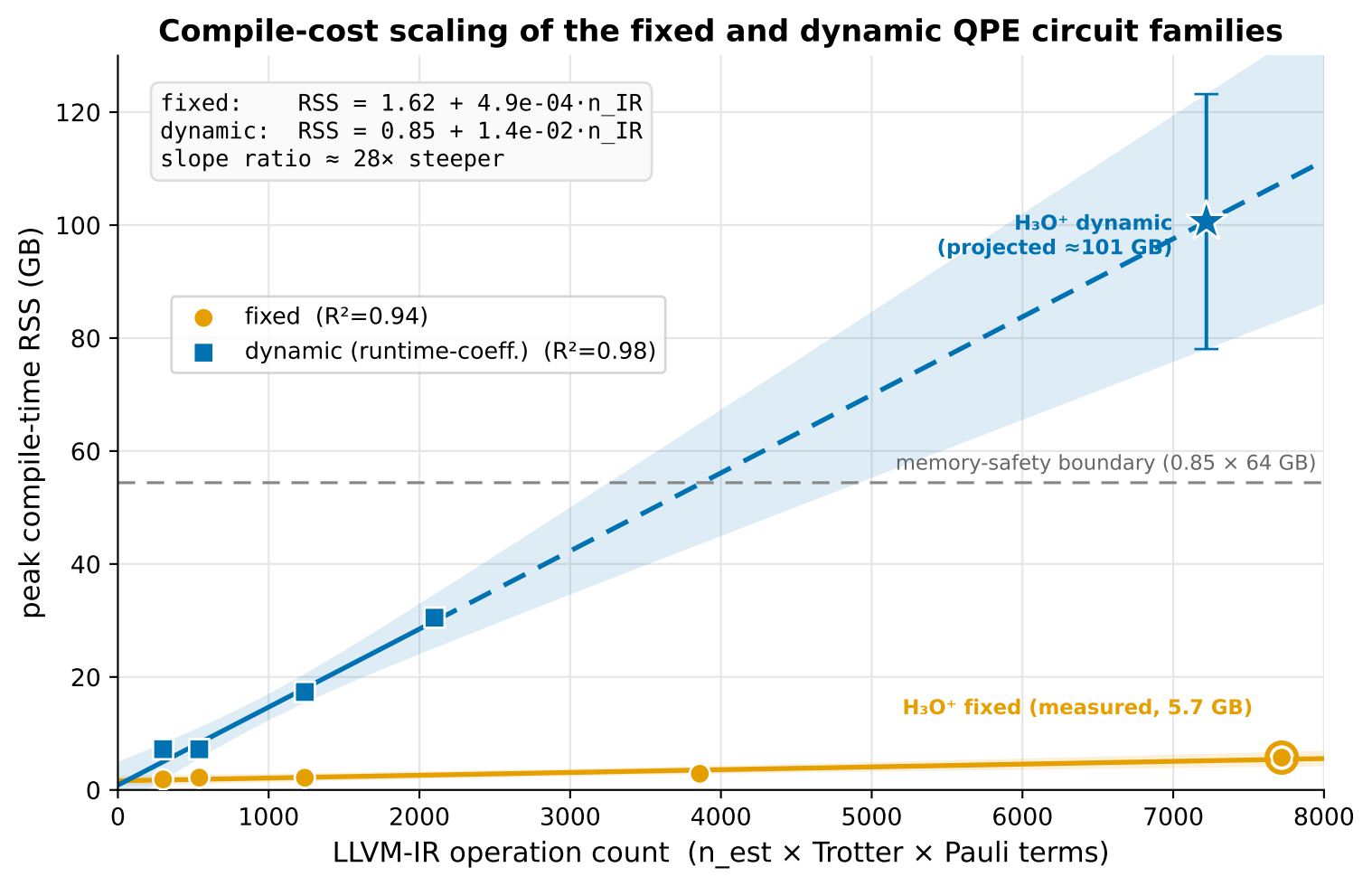

Compile-cost scaling of the fixed and dynamic QPE circuit families
fixed: RSS = 1.62 + 4.9e-04·n_IR
dynamic: RSS = 0.85 + 1.4e-02·n_IR
slope ratio ≈ 28× steeper
fixed (R²=0.94)
dynamic (runtime-coeff.) (R²=0.98)
H₃O⁺ dynamic
(projected ≈101 GB)
memory-safety boundary (0.85 × 64 GB)
H₃O⁺ fixed (measured, 5.7 GB)
peak compile-time RSS (GB)
LLVM-IR operation count (n_est × Trotter × Pauli terms)
0
20
40
60
80
100
120
0
1000
2000
3000
4000
5000
6000
7000
8000